🧪 Exercise: Firewall Setup with Ansible-Automation

Here's a complete **Ansible playbook** that configures a basic **UFW firewall** for a Kubernetes cluster with:

📝 Objective

You will learn to:

- Install and enable the UFW firewall via Ansible on:
- One **control-plane node**
- One or more **worker nodes**
- assign roles (control-plane or worker) using **host groups** in your Ansible inventory.

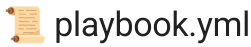
📁 Directory structure (recommended)

```
k8s-firewall/
├── inventory.ini
├── playbook.yml
```

📘 inventory.ini

```
[control_plane]
master1 ansible_host=192.168.0.10

[worker]
worker1 ansible_host=192.168.0.11
worker2 ansible_host=192.168.0.12
```


playbook.yml

```
---
- name: Configure UFW firewall for Kubernetes nodes
  hosts: all
  become: true
  vars:
    internal_subnet: "192.168.0.0/24"

  tasks:
    - name: Install UFW
      apt:
        name: ufw
        state: present
        update_cache: yes

    - name: Set default policies
      ufw:
        direction: incoming
        policy: deny

    - name: Allow outgoing traffic
      ufw:
        direction: outgoing
        policy: allow

    - name: Allow SSH
      ufw:
        rule: allow
        port: 22
        proto: tcp

- name: Configure control-plane firewall rules
  hosts: control_plane
  become: true
  tasks:
    - name: Allow Kubernetes API server
      ufw:
        rule: allow
        port: 6443
        proto: tcp

    - name: Allow etcd ports
      ufw:
        rule: allow
        port: "{{ item }}"
        proto: tcp
      loop: [2379, 2380]

    - name: Allow kubelet API
      ufw:
        rule: allow
        port: 10250
        proto: tcp
```

```
- name: Allow scheduler and controller-manager
      ufw:
        rule: allow
        port: "{{ item }}"
        proto: tcp
      loop: [10257, 10259]

    - name: Enable UFW
      ufw:
        state: enabled

- name: Configure worker node firewall rules
  hosts: worker
  become: true
  tasks:
    - name: Allow kubelet API
      ufw:
        rule: allow
        port: 10250
        proto: tcp

    - name: Allow NodePort range
      ufw:
        rule: allow
        port: "30000:32767"
        proto: tcp

    - name: Enable UFW
      ufw:
        state: enabled
```

▶️ Run the playbook

```
ansible-playbook -i inventory.ini playbook.yml
```

> 🔐 Use `--ask-become-pass` if you don't use passwordless sudo.

✅ Outcome

- UFW will be enabled on all nodes
- Only the necessary Kubernetes ports will be open
- Everything else will be blocked

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